

Bridging the Gap between Hybrid and Fuel Cars

Mariam Taj Muhammad



Table of contents

01

Introduction

02

**Problem
Statement**

03

**Data
Overview**

04

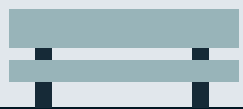
Analysis

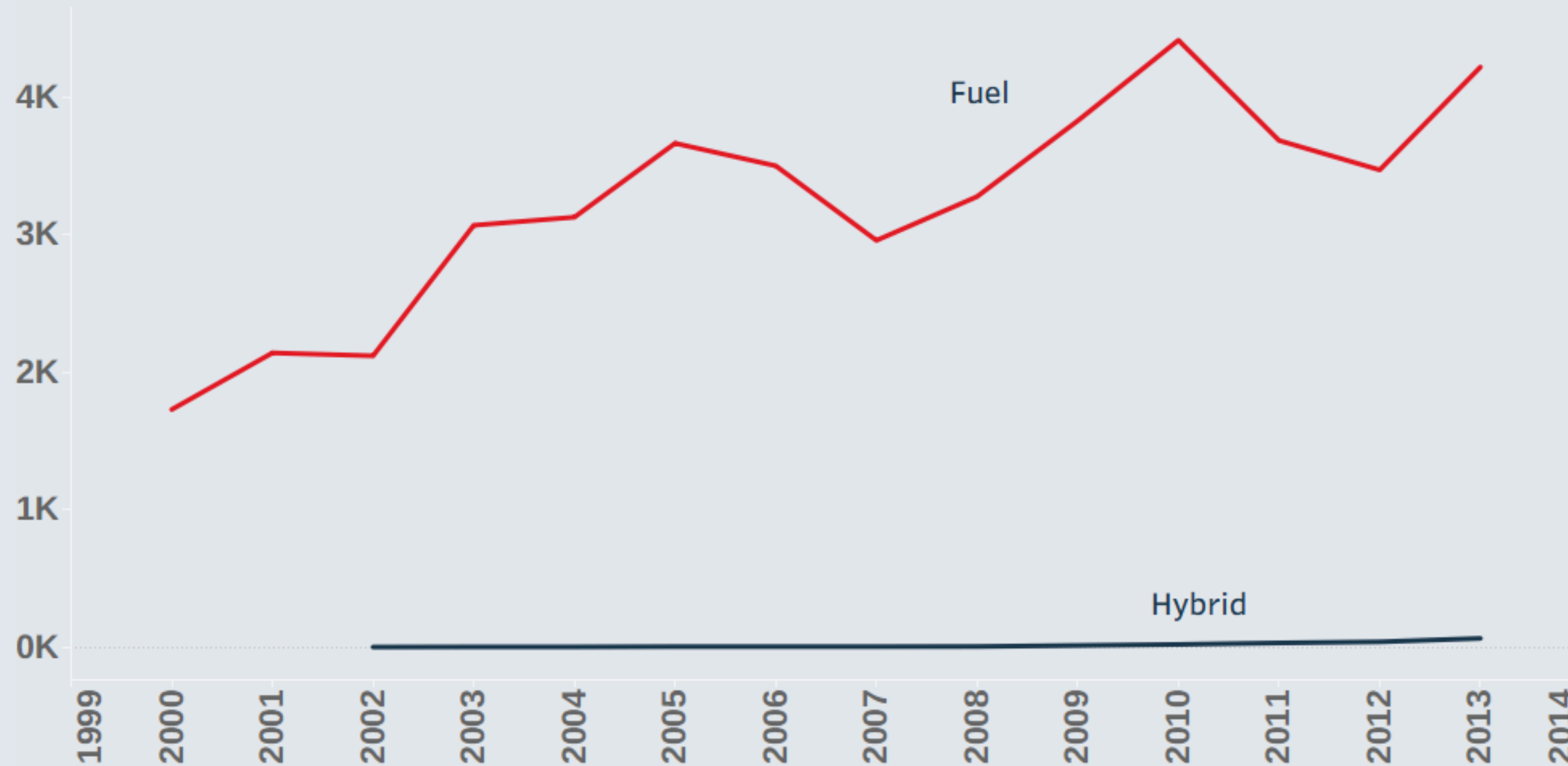
05

Recommendation

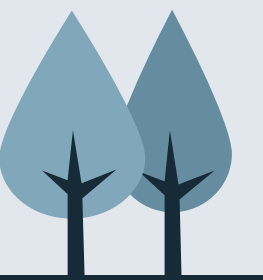
Introduction

The transportation sector, as one of the largest contributors to greenhouse gas emissions, has become a central focus of policy initiatives aimed at promoting **cleaner** and more **sustainable** alternatives.





Fuel-powered cars consistently outnumbered hybrid vehicles by a factor of **44** between 2000 and 2013 in the UK.



Aim

To identify the **key barriers** that limit the uptake of **hybrid** cars compared to conventional **fuel** cars and assess the **implications** on emission goals of the UK.



Objectives

1. To compare the **economic performance** of **hybrid** and **fuel**-powered cars in the UK from 2000 to 2013.
2. To evaluate **consumer awareness** and **perceptions** of **hybrid** technology.
3. To assess the impact of **technological** and **performance**-related factors on the adoption of **hybrid** vehicles.
4. To examine the role of **infrastructure** in shaping the uptake of **hybrid** cars.
5. To analyze how the continued dominance of **fuel**-powered cars affected the UK's progress toward achieving cleaner transportation and **emission reduction goals**.

UK Government

2000 - 2013

**Fuel Types:
Fuel/ Hybrid/ **Electric****

Data Overview

13

Countries

45K

Cars

61

Manufacturers

12

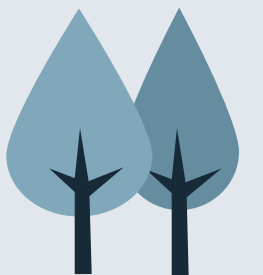
**Market
Segments**

Why do consumers *resist* **hybrid** cars?

Price

Perception

Performance

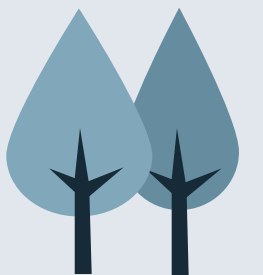


Why do consumers *resist* **hybrid** cars?

Price

Perception

Performance



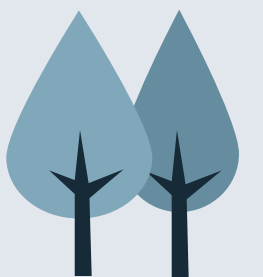
Price

Base_price * Manufacturer_multiplier + Emission_tax

Base_price: Based on **manufacturer market segment** (luxury, mass-market, sports, etc..) factored to **2013**.

Manufacturer_multiplier: Manufacturer **rank** per market segment.

Emission_tax: Based on **CO₂** emissions.



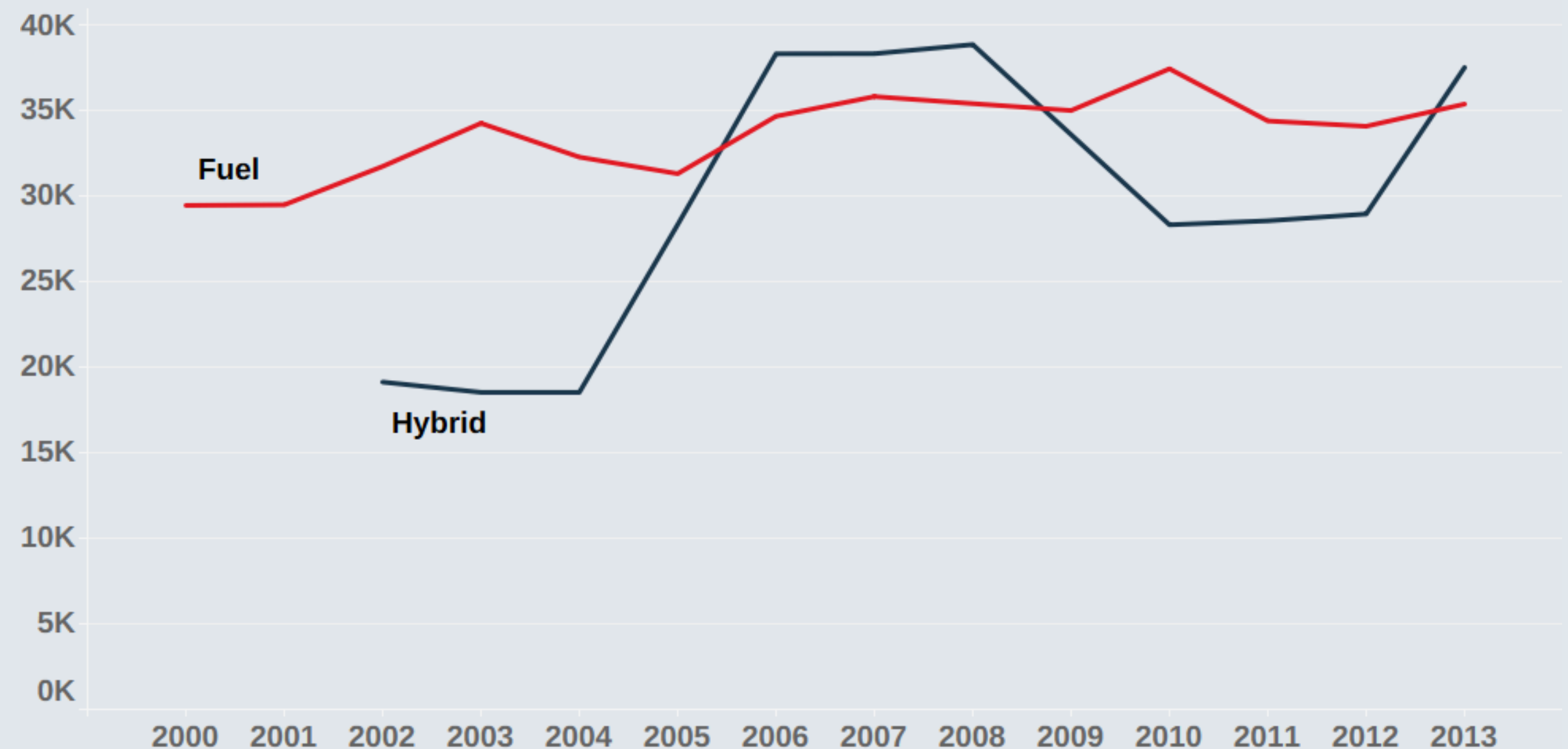
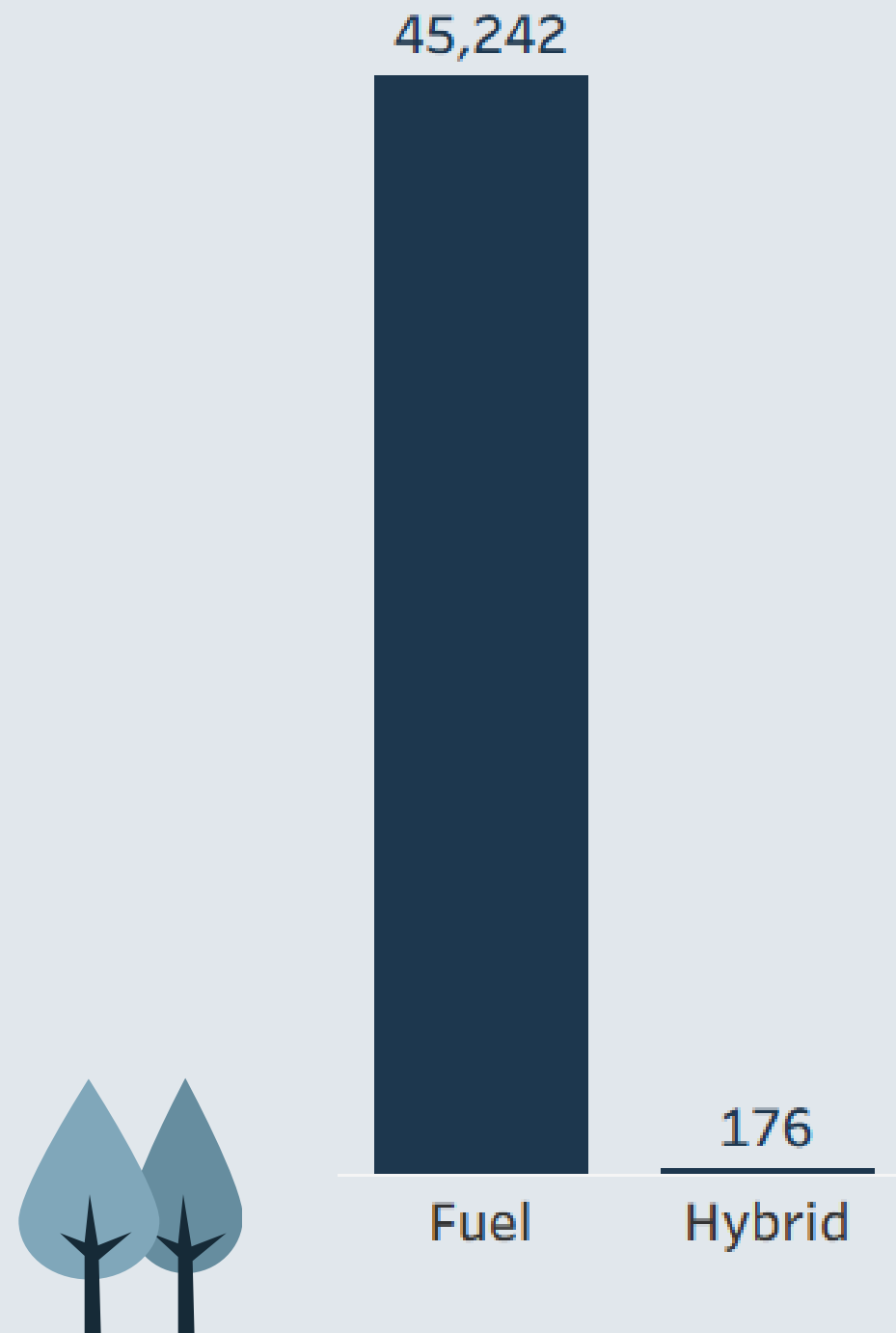
Tax Bands

Band	CO2 emission (g/km)	Tax
A	0-100	£ 20
B	101-110	£ 20
C	111-120	£ 35
D	121-130	£ 165
E	131-140	£ 195
F	141-150	£ 215
G	151-165	£ 265
H	166-175	£ 315
I	176-185	£ 345
J	186-200	£ 395
K*	201-225	£ 430
L	226-255	£ 735
M	251-Above	£ 760



Average Car Prices over the Years

Total no. of cars
(2000-2013)



Hybrid to Fuel Price Ratio

0.989

Hybrids are slightly cheaper.

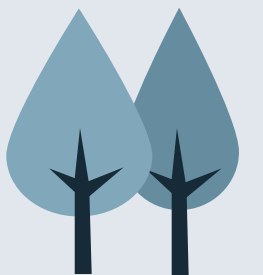


Why do consumers *resist* hybrid cars?

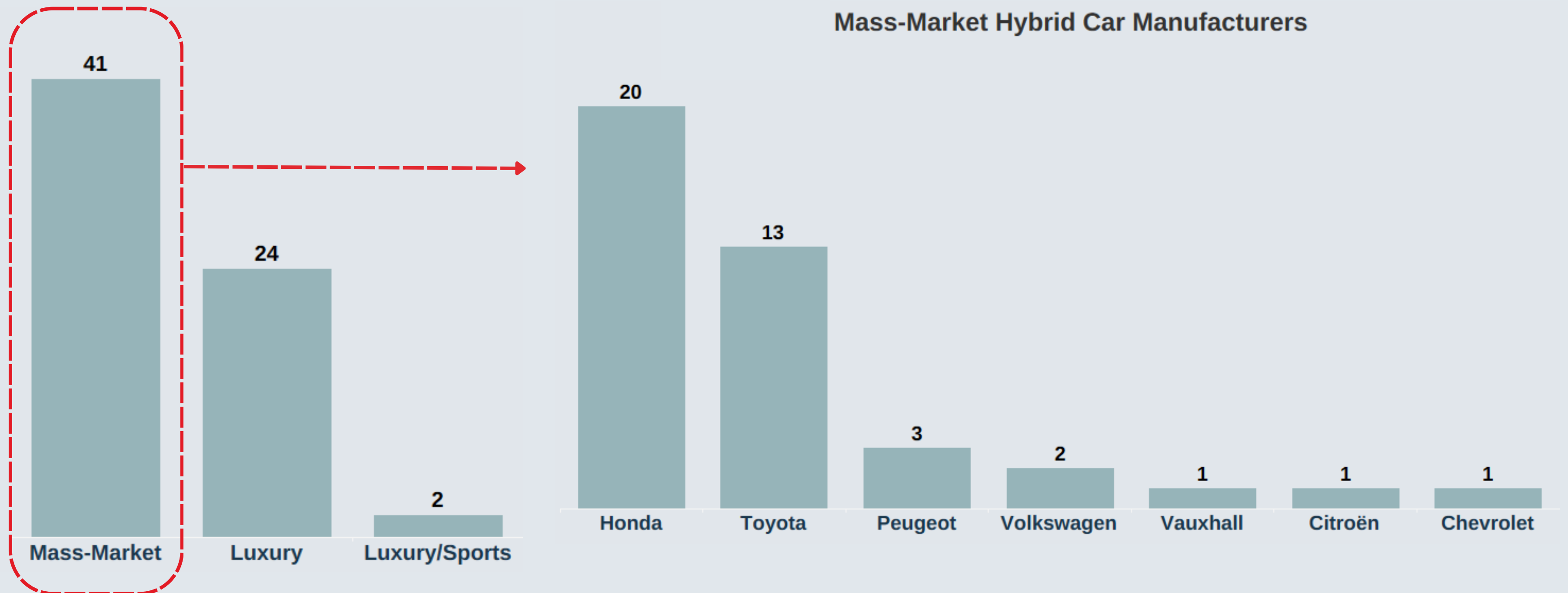
Price

Perception

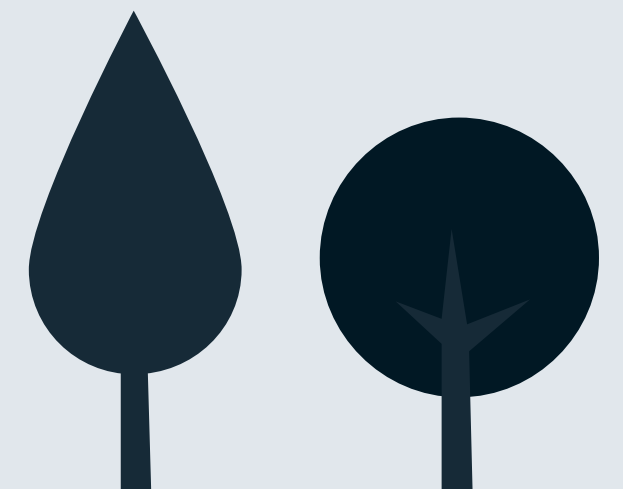
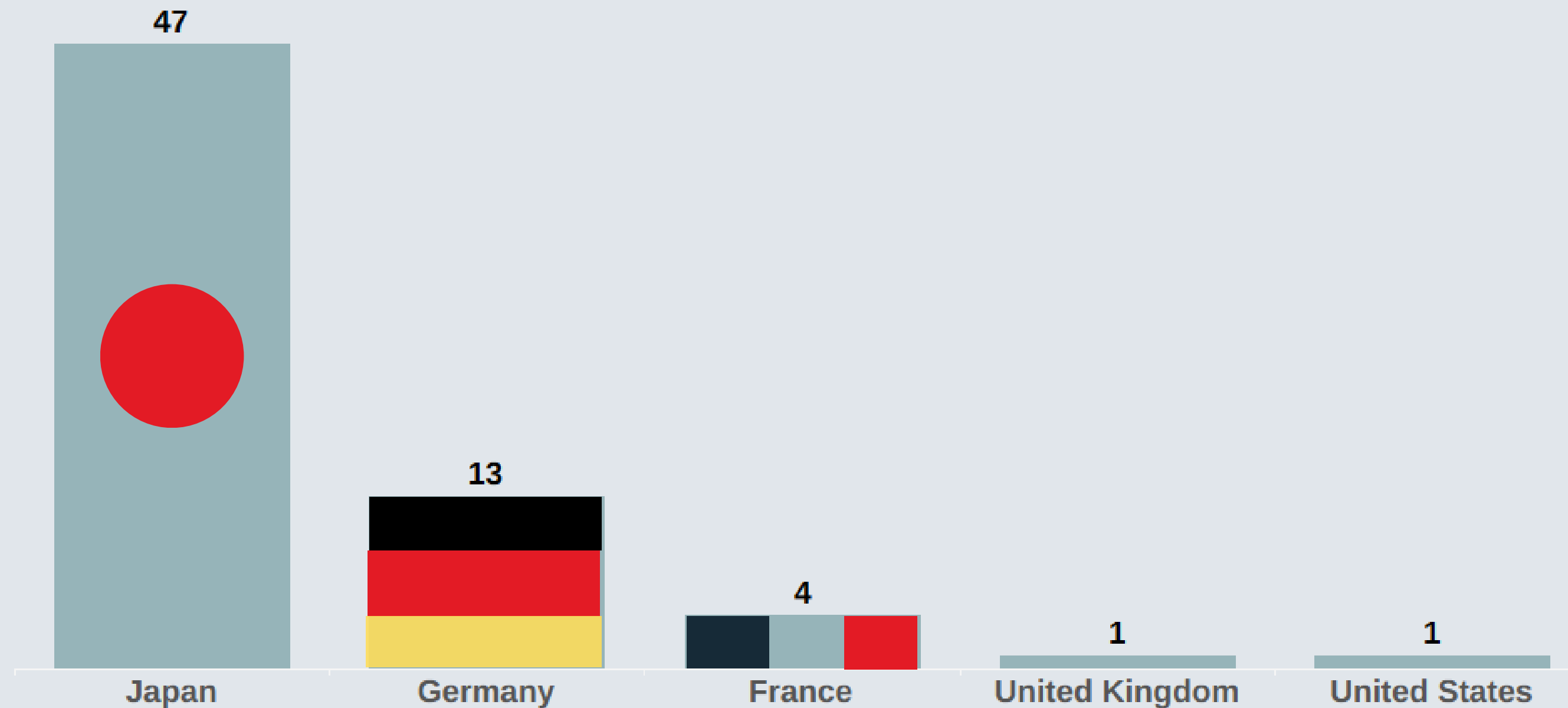
Performance



Market Segments of 2000-2013 Manufactured Hybrid Cars in the UK



Top Hybrid Car Manufacturing Countries 2000-2013

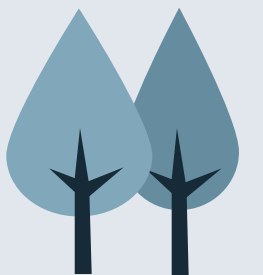


Why do consumers *resist* **hybrid** cars?

Price

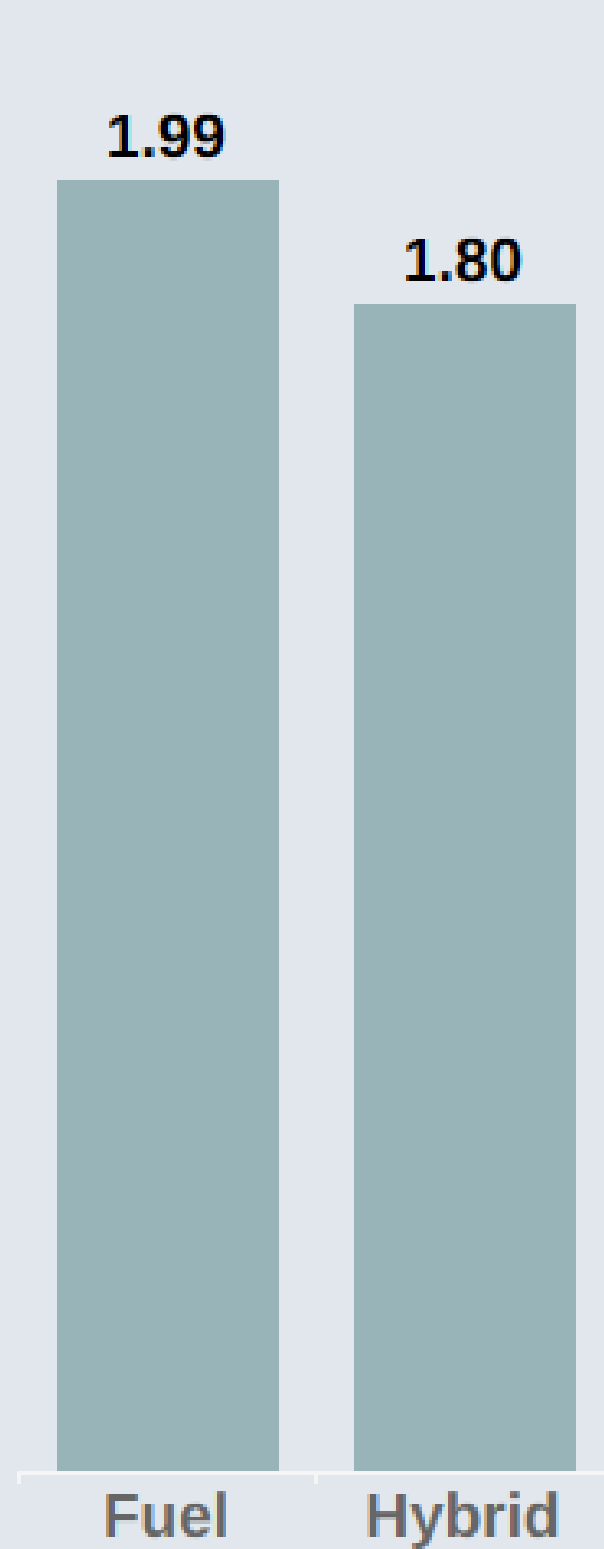
Perception

Performance

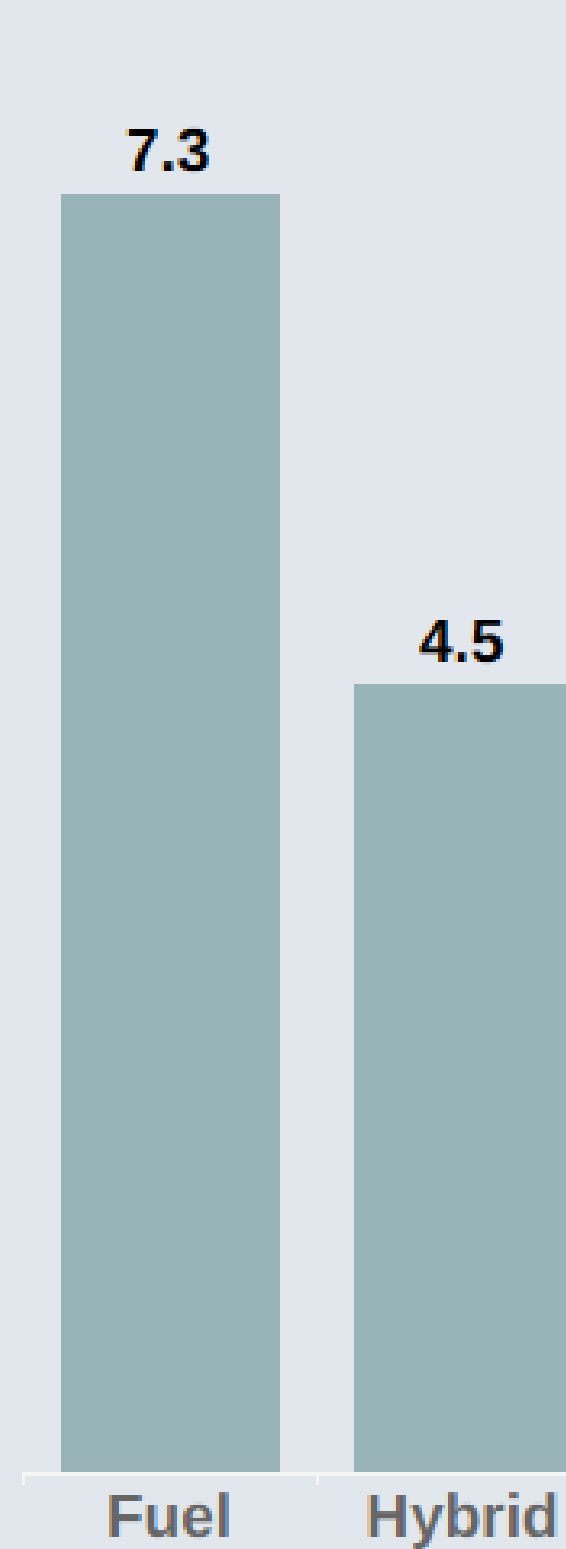


*Median

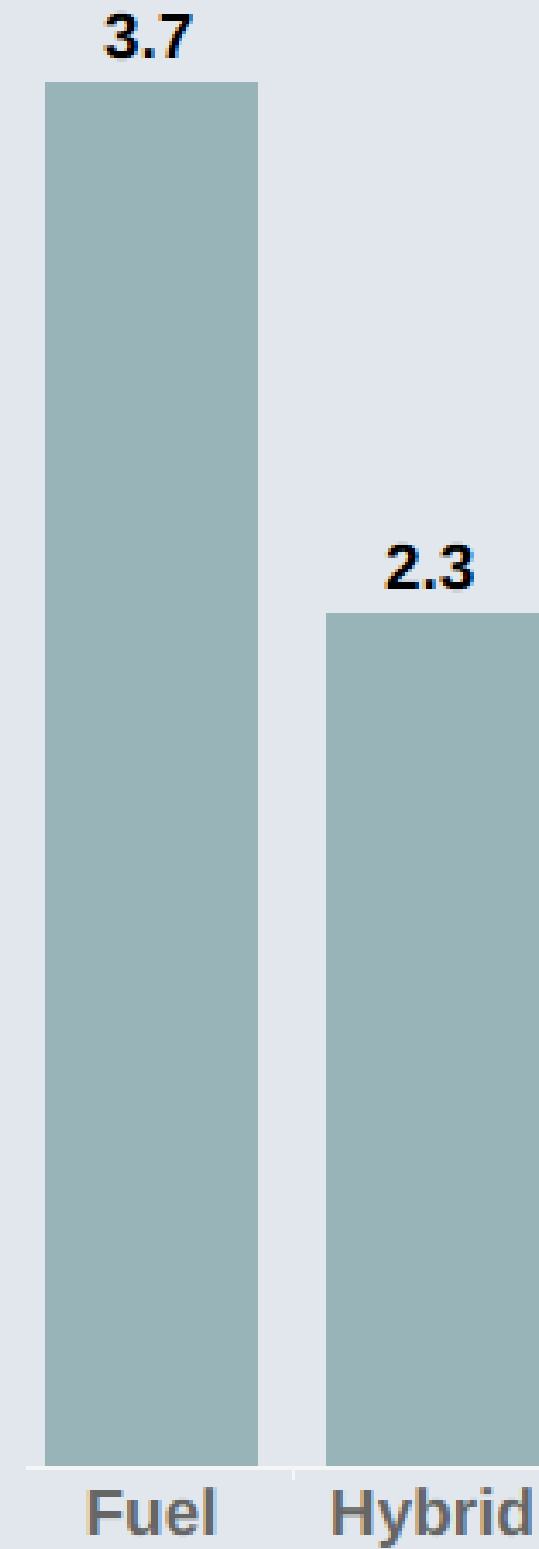
Engine Capacity [L]*



Fuel Consumption [L/100km]*



Aerodynamics Factor



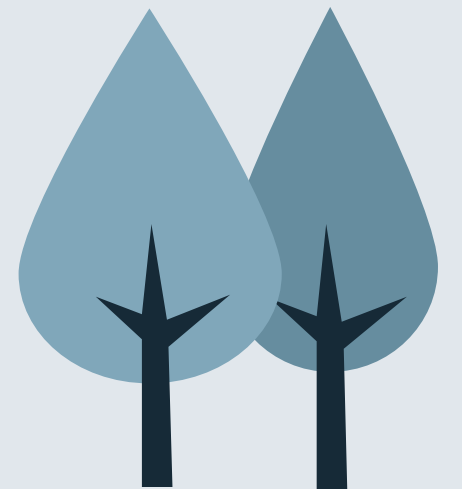
Noise [dB(A)]*



Efficiency Gap

2.57

Hybrids consume less fuel.



Performance Gap

1.793

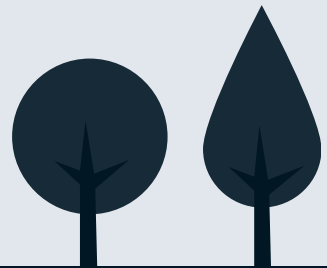
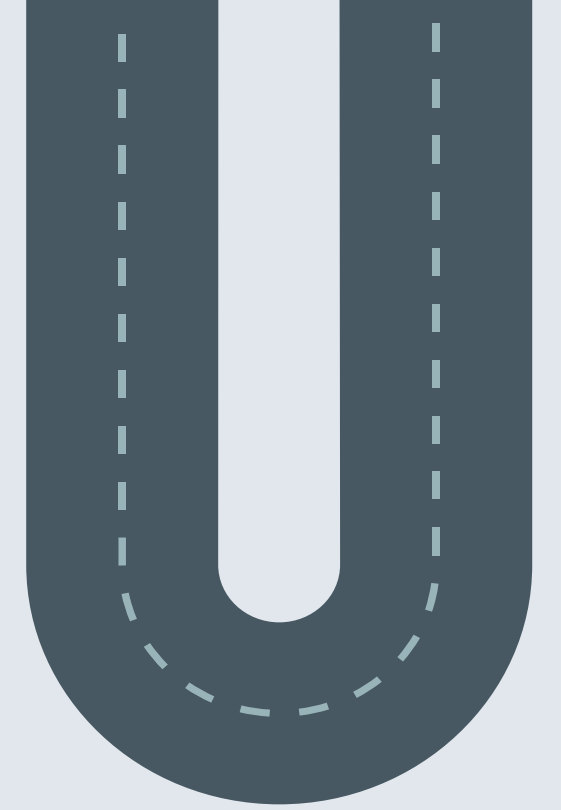
Fuel cars have 9% better performance.

$$PerformanceIndex = Capacity^{(f)}$$

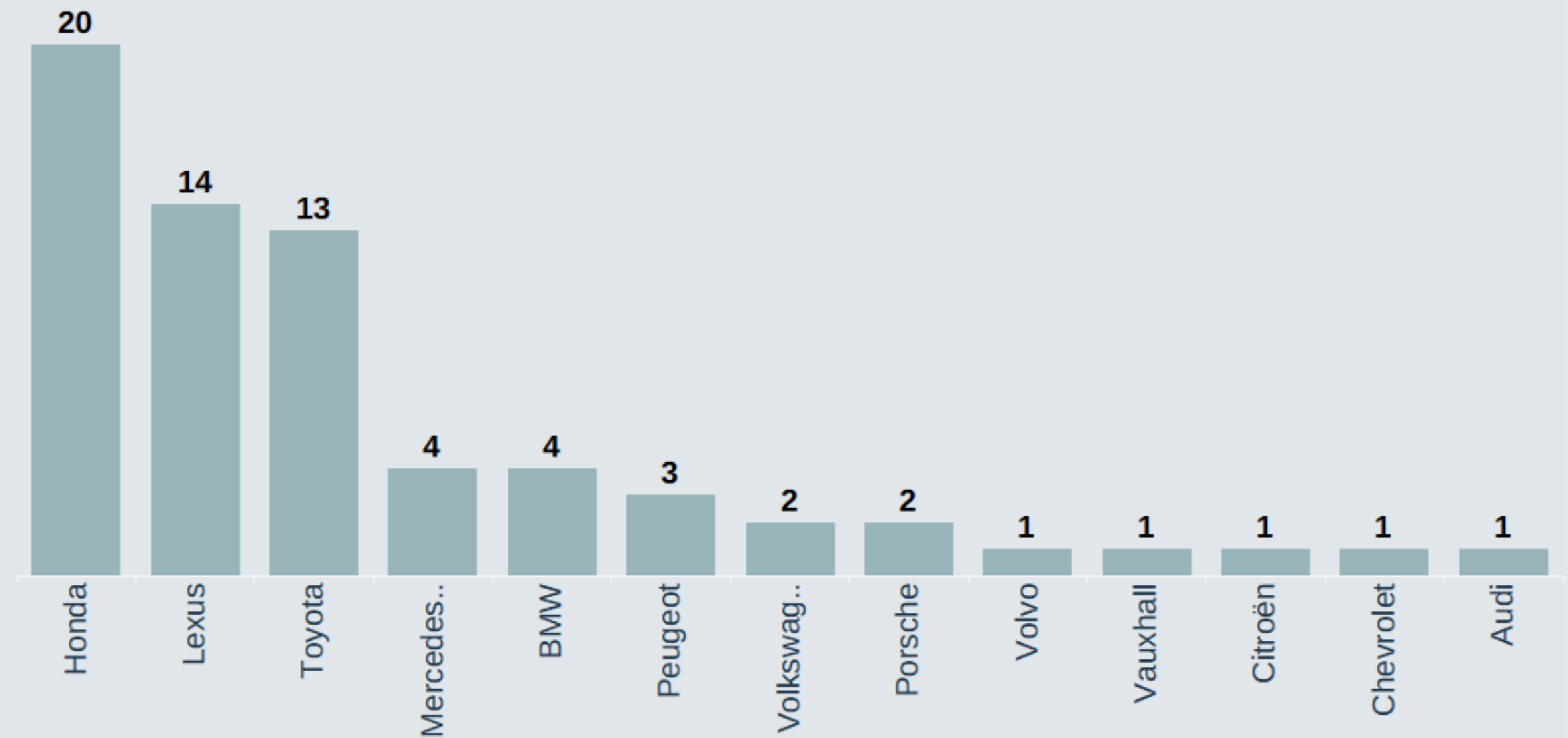
$$0.7 > \mathbf{f} < 1.0$$

$$\mathbf{f} \approx 0.85$$

Infrastructure and Clean Transition



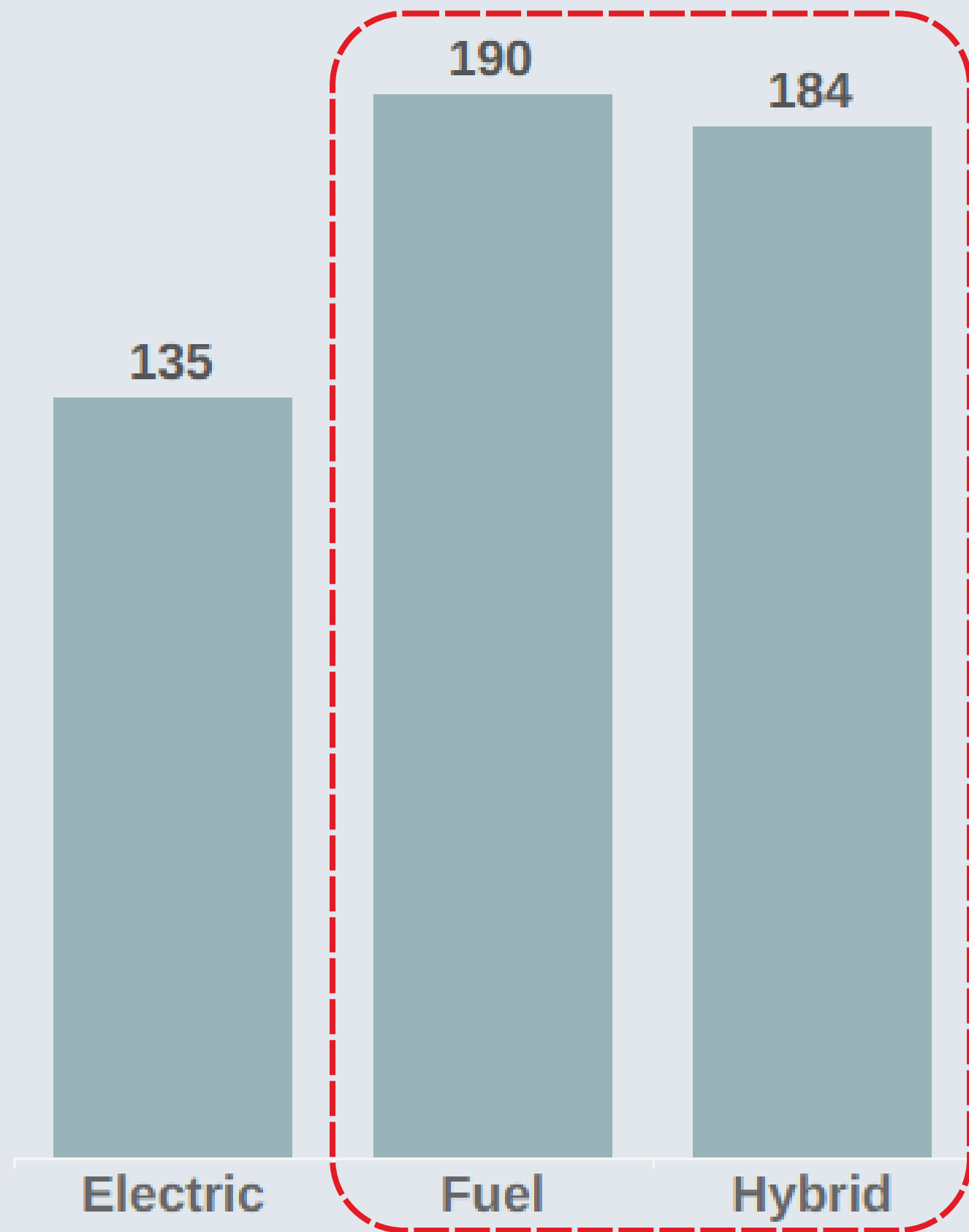
Hybrid Cars Distribution by Manufacturer



Hybrids in 2000-2013 were mainly Self-charging

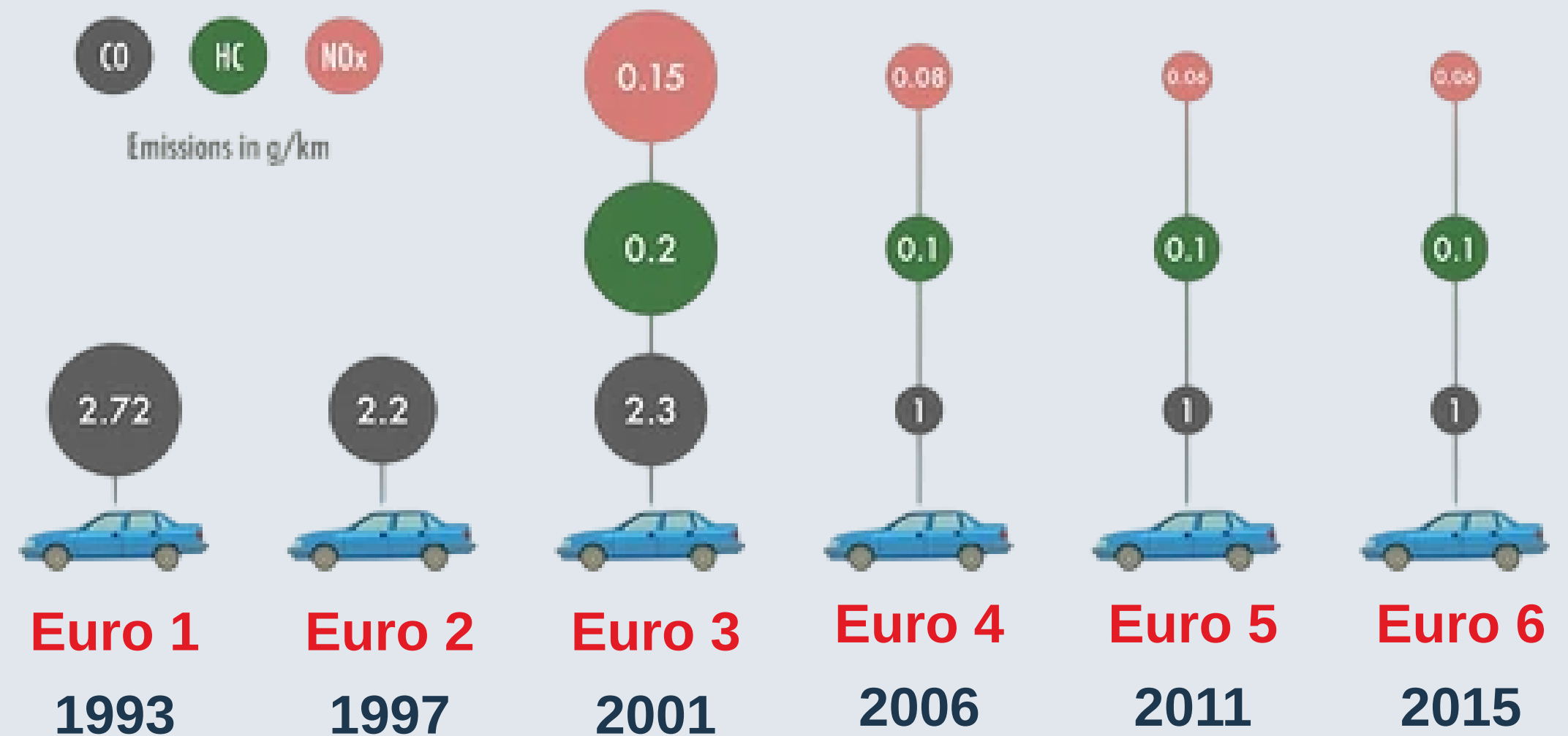


Average CO₂ Emissions Over the Years

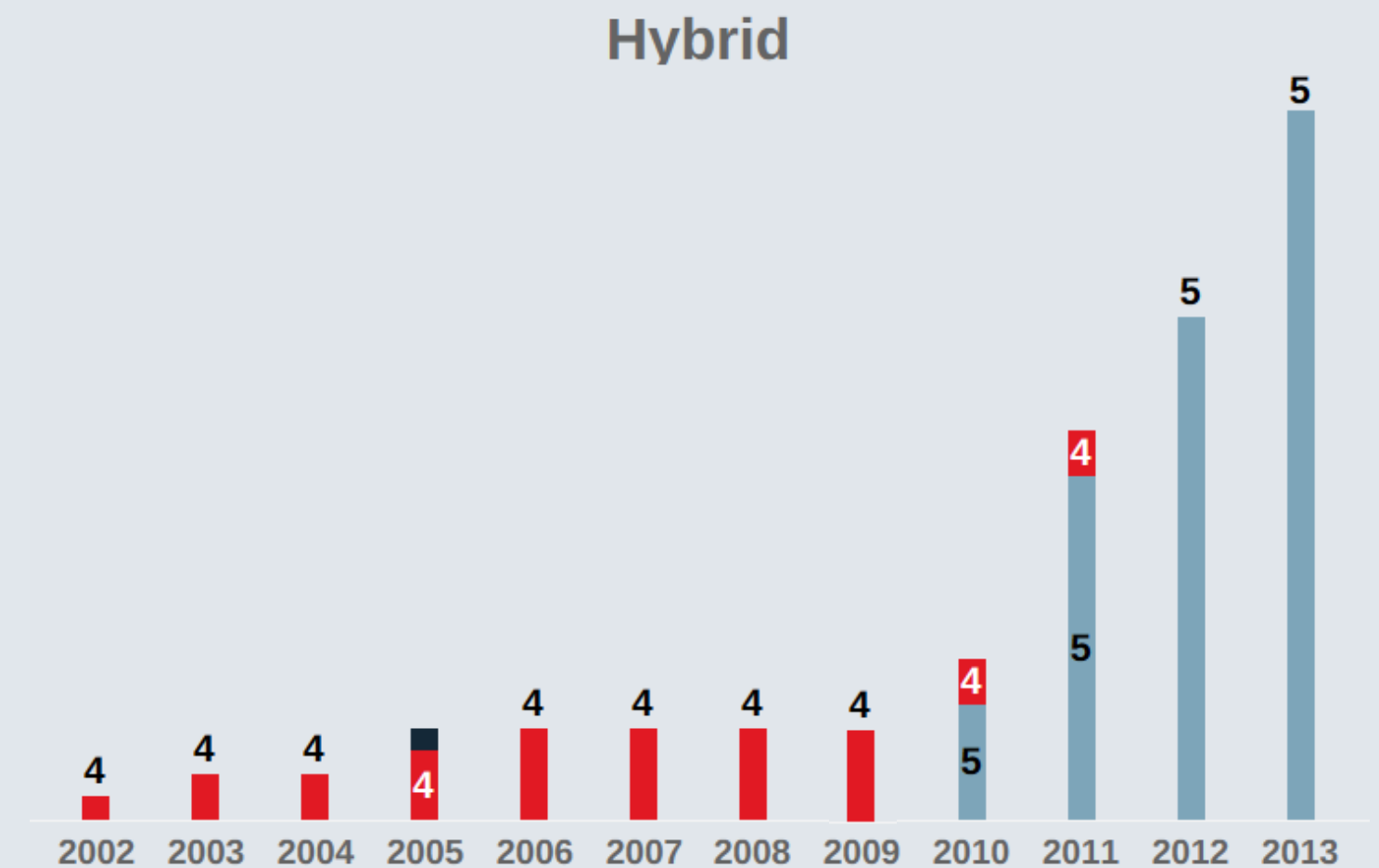
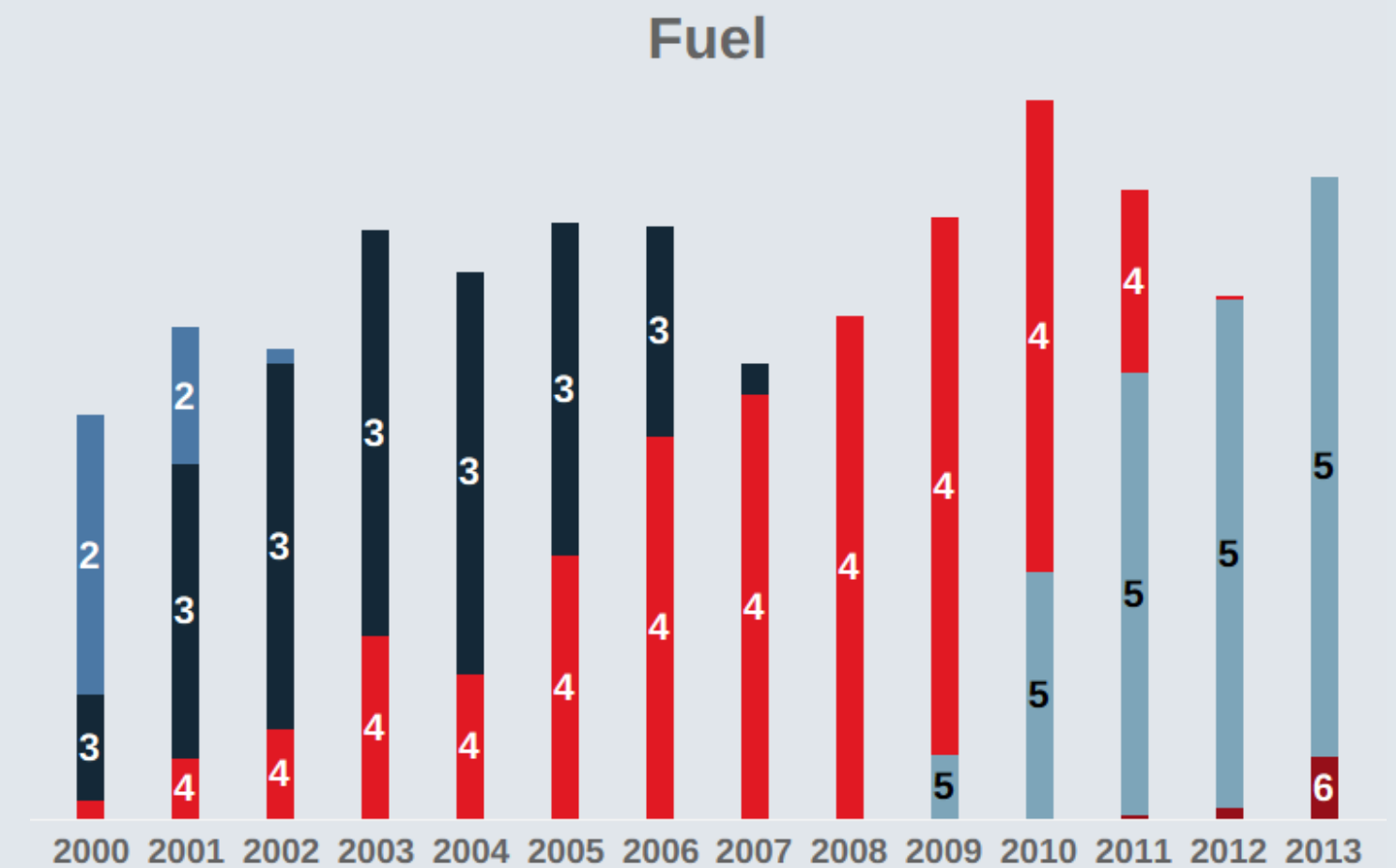
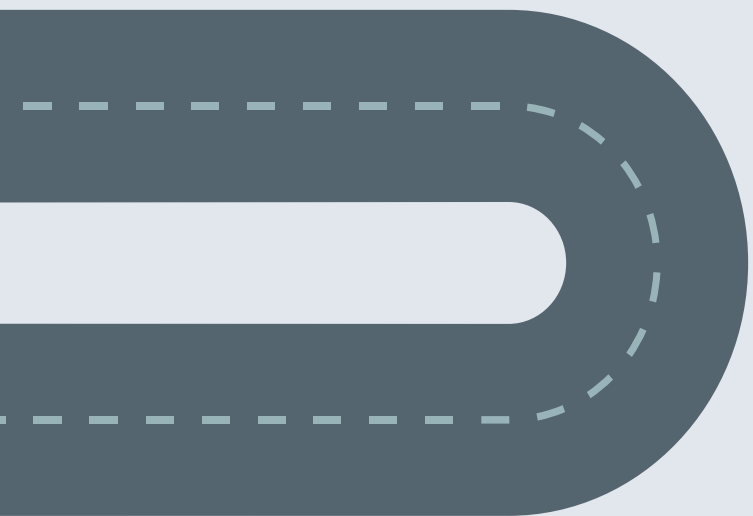


Euro Emission Standards

European Union
(EU) regulations
limiting harmful
exhaust pollutants
from new vehicles.



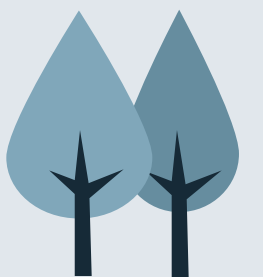
Euro Emission Standards compliance Over the Years



Conclusion

Fuel cars remained dominant in the UK (2000–2013), outnumbering **hybrids** **44:1** despite better efficiency and lower emissions.

High costs, limited infrastructure, and weak consumer confidence slowed the transition to cleaner transport.

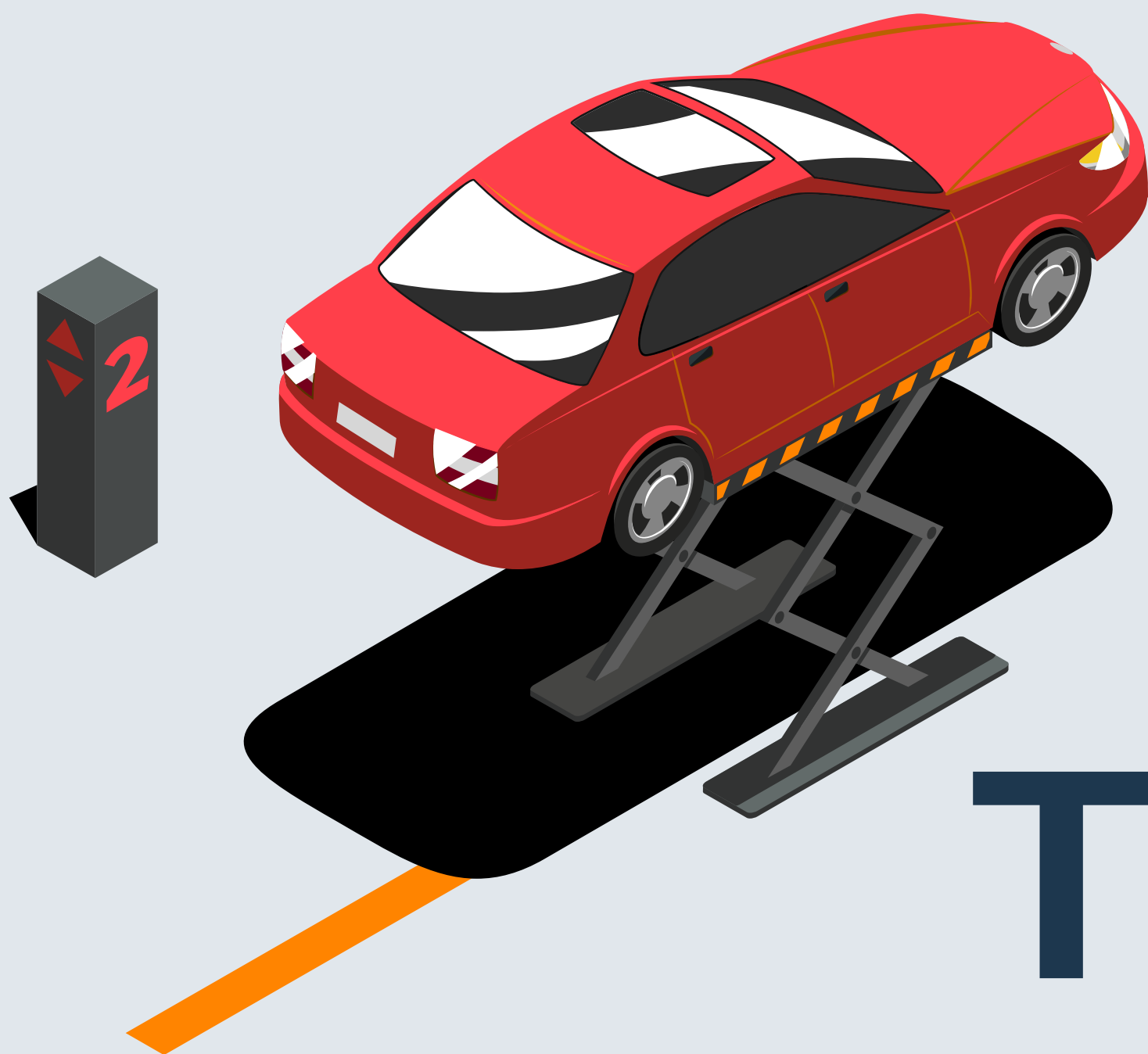


Recommendation

- Strengthen incentives and expand hybrid infrastructure.
- Raise consumer awareness and trust in hybrid technology.
- Broaden manufacturer participation across all market segments.
- Enforce stricter emission and efficiency standards.

What Has Been Implemented (2025)

- Strengthen incentives and expand hybrid infrastructure.
- Raise consumer awareness and trust in hybrid technology.
- Broaden manufacturer participation across all market segments.
- Enforce stricter emission and efficiency standards.



Thank you!